

Mechatronics System Design

Assignment-1

Total Marks: 100

Due Date: 31-03-2026

Late Submission: Late submission will receive a penalty for each additional day past the due date.

Instructions:

1. This assignment must be completed individually.
 2. Submit the script, images, and graphs, along with the program outputs, in their respective folders provided in the zip file.
 3. Prepare a single PDF report covering all questions.
 4. Submit one assignment per team as a zip file named `roll_number.zip`.
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Overview

In this assignment, you will design a four-bar linkage mechanism using Fusion 360. This practical exercise will help you understand the principles of mechanism design and kinematic analysis in the context of mechatronic systems.

Part-1: Four-Bar Linkage Mechanism Design and ROS2 Simulation

1.1 CAD Design in Fusion 360

Objective

Design a four-bar coupling mechanism of crank-rocker-rocker configuration in Fusion 360.

Design Requirements

1. Design a four-bar coupling mechanism of crank-rocker-rocker configuration.
2. Each bar must be created as a **separate component** in Fusion 360.
3. Components must be properly connected using **revolute joints**.
4. The mechanism must satisfy **Grashof's condition** to ensure proper motion.
5. The design should be functional and demonstrate smooth motion throughout the crank's rotation.

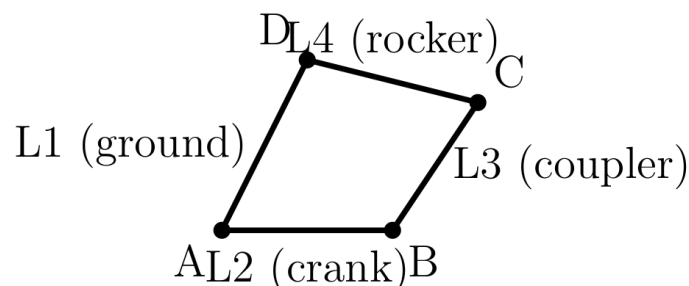


Figure 1: Schematic representation of a four-bar linkage mechanism.

Design Process Guidelines

1. Create a new design in Fusion 360.
2. Design each link as a separate component:
 - **Ground link (L_1):** Fixed to the global coordinate system.
 - **Input link/Crank (L_2):** Connected to the ground at one end.
 - **Coupler link (L_3):** Connected between the crank and the rocker.
 - **Output link/Rocker (L_4):** Connected to the ground at one end.
3. Use revolute joints to connect the components:
 - **Joint A:** Ground (L_1) to Crank (L_2).
 - **Joint B:** Crank (L_2) to Coupler (L_3).
 - **Joint C:** Coupler (L_3) to Rocker (L_4).
 - **Joint D:** Rocker (L_4) to Ground (L_1).
4. Test the mechanism by rotating the crank through a full 360° rotation.
5. Verify that the mechanism moves as expected and does not lock up.

Important Notes for CAD Design

- The crank must be able to complete a full 360° rotation without the mechanism binding.
- Ensure your design follows **Grashof's condition**: $S + L \leq P + Q$, where:
 - S is the shortest link length,
 - L is the longest link length,
 - P and Q are the remaining link lengths.
- For optimal visualization in your video, consider adding colors to differentiate the links.
- You may add additional features to your design for improved functionality or aesthetics, but the basic four-bar linkage must function as required.

1.2 URDF Generation and ROS2 Simulation

Objective

Convert the CAD model from Part 1.1 into a URDF (Unified Robot Description Format) file, import it into a ROS2 package, and simulate the mechanism with oscillatory motion at Joint B.

What is URDF?

URDF is an XML format used in ROS (Robot Operating System) to describe robot models, including their kinematic structure, visual appearance, collision properties, and joint behaviors. Converting your CAD model to URDF allows you to simulate and control your mechanism in a ROS2 environment.

Design Requirements

1. Export URDF from Fusion 360:

- Use the `fusion2urdf` plugin to convert your Fusion 360 model to URDF format.
- Ensure all components are properly defined as separate components (not bodies).
- Verify that all revolute joints are correctly exported with proper parent-child relationships.

2. Create ROS2 Package:

- Create a new ROS2 package to contain your URDF model.
- Set up proper directory structure with `urdf`, `launch`, and `config` folders.
- Include necessary dependencies (`robot_state_publisher`, `joint_state_publisher`).

3. Implement Joint B Oscillatory Motion:

- Write a ROS2 node (Python or C++) that publishes joint states.
- Implement oscillatory motion for **Joint B** with the following specifications:
 - **Range:** -30° to $+30^\circ$ (or $-\pi/6$ to $+\pi/6$ radians).
 - **Motion Type:** Sinusoidal or triangular wave oscillation.
 - **Frequency:** Choose an appropriate frequency (e.g., 0.5 Hz–1.0 Hz).

4. Visualization in RViz2:

- Create a launch file to start RViz2 with your robot model.
- Configure RViz2 to properly display the mechanism.
- Visualize the oscillatory motion of Joint B and observe the resulting motion of the entire mechanism.

Step-by-Step You Should Follow

- Step 1: Prepare Fusion 360 Model for URDF Export
- Step 2: Install and Use `fusion2urdf` Plugin
- Step 3: Create ROS2 Package
- Step 4: Write Joint State Publisher Node
- Step 5: Create Launch File
- Step 6: Build and Run

Submission Requirements

You must submit the following:

1. The complete Fusion 360 design file (`.f3d`, `.step`, `.stl` or `.dae`) of your four-bar linkage mechanism.
2. A video (maximum 30 seconds) showing the movement of your mechanism in Fusion 360 with the crank rotating a full 360° .
3. Exported URDF file with all associated mesh files (`.stl` or `.dae`).

4. Complete ROS2 package.
5. A screen recording or video (maximum 1 minute) showing:
 - RViz2 visualization of your mechanism,
 - Joint B oscillating in the range $[-30^\circ, +30^\circ]$,
 - The resulting motion of the entire four-bar linkage.
6. A brief report (1–2 pages) explaining:
 - Your design choices and link dimensions,

Resources for Part 1

Fusion 360 Resources

- Fusion 360 Linkage Mechanisms Tutorial: <https://www.youtube.com/watch?v=Ct6RUt9LhTQ>

CAD to URDF Conversion

- fusion2urdf GitHub Repository: <https://github.com/syuntoku14/fusion2urdf>
- Fusion2URDF Installation Tutorial: https://www.youtube.com/watch?v=TitHYg-5_j8
- Fusion 360 to URDF Course: <https://www.youtube.com/watch?v=EuV1rsfjBJo>

ROS2 and URDF Resources

- ROS2 Official URDF Documentation: <https://docs.ros.org/en/humble/Tutorials/Intermediate/URDF/URDF-Main.html>
- Creating URDF and Launch Files in ROS2: <https://aleksandarhaber.com/how-to-create-urdf-and-launch-files-in-ros2/>
- ROS2 URDF Tutorial Video: <https://www.youtube.com/watch?v=jDsb8xEdbKM>
- ArmBot Fusion360 to URDF in ROS2: https://www.youtube.com/watch?v=_ZFo6wPXjeQ

Joint Control and Simulation

- ROS2 Control Documentation: https://control.ros.org/rolling/doc/ros2_control/hardware_interface/doc/joints_userdoc.html
- Publishing Joint States in ROS2: <https://www.youtube.com/watch?v=p5LVg6Fre18>

End of Assignment